

## ACTIVITY

**TOOL:R Programming**

**RegisterNo:192424335**

**Name: M. Rishitha**

**Date:24-07-2026**

**COURSE: Data Handling and Visualization**

**Code: DSA0604**

**EMPLOYEE SALARY DATASET**

**1.AMOUNTS**

**BAR PLOT VISUALIZATION**

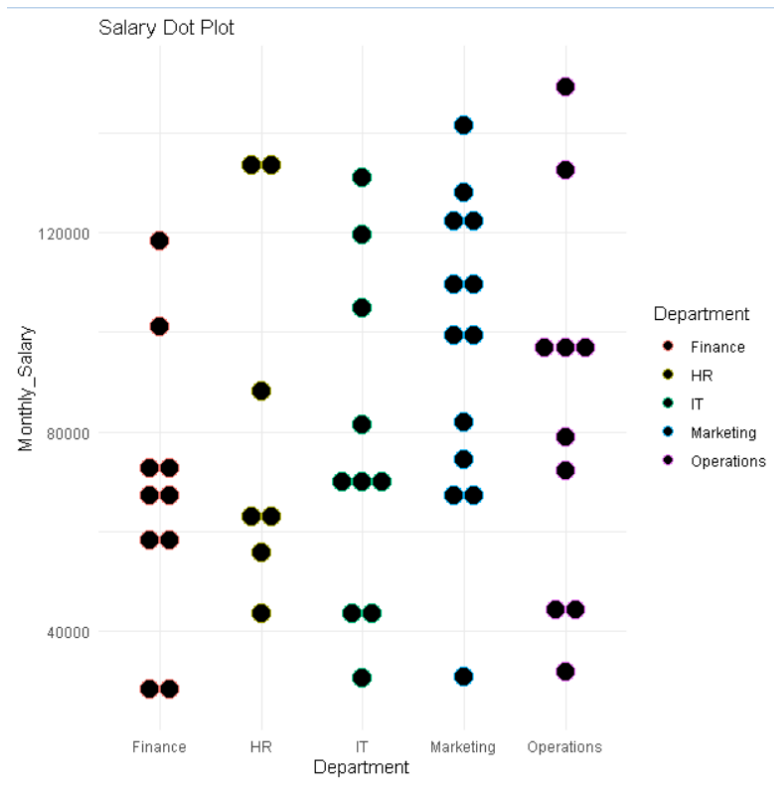
```
ggplot(employee,  
  aes(x = Department,  
    y = Monthly_Salary,  
    fill = Department)) +  
stat_summary(fun = mean, geom = "bar") +  
labs(title = "Average Salary by Department",  
  x = "Department",  
  y = "Average Monthly Salary") +  
theme_minimal()
```



## 2.DOT PLOT VISUALIZATION

### R PROGRAM

```
ggplot(employee,  
  aes(x = Department,  
    y = Monthly_Salary,  
    color = Department)) +  
geom_dotplot(binaxis = "y",  
  stackdir = "center") +  
labs(title = "Salary Dot Plot") +  
theme_minimal()
```



### 3.HEAT MAP

#### R PROGRAM

```
library(dplyr)
```

```
heat_data <- employee %>%
```

```
  group_by(Department, Education_Level) %>%
```

```
  summarise(Average_Salary = mean(Monthly_Salary),
    .groups = "drop")
```

```
ggplot(heat_data,
```

```
  aes(x = Department,
```

```
    y = Education_Level,
```

```
    fill = Average_Salary)) +
```

```
geom_tile(color = "white") +
```

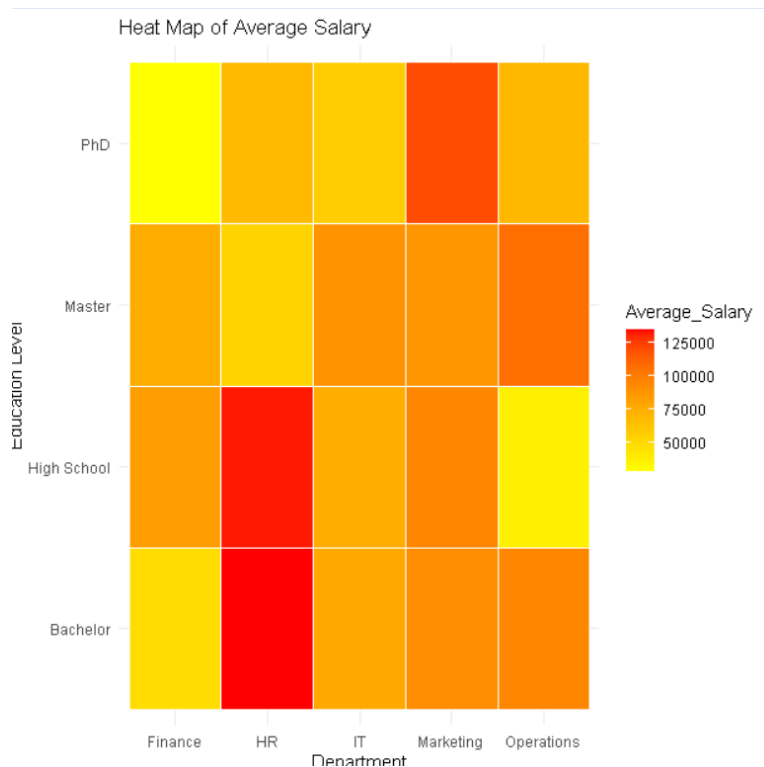
```
scale_fill_gradient(low = "yellow",
```

```
  high = "red") +
```

```
labs(title = "Heat Map of Average Salary",
```

```
  x = "Department",
```

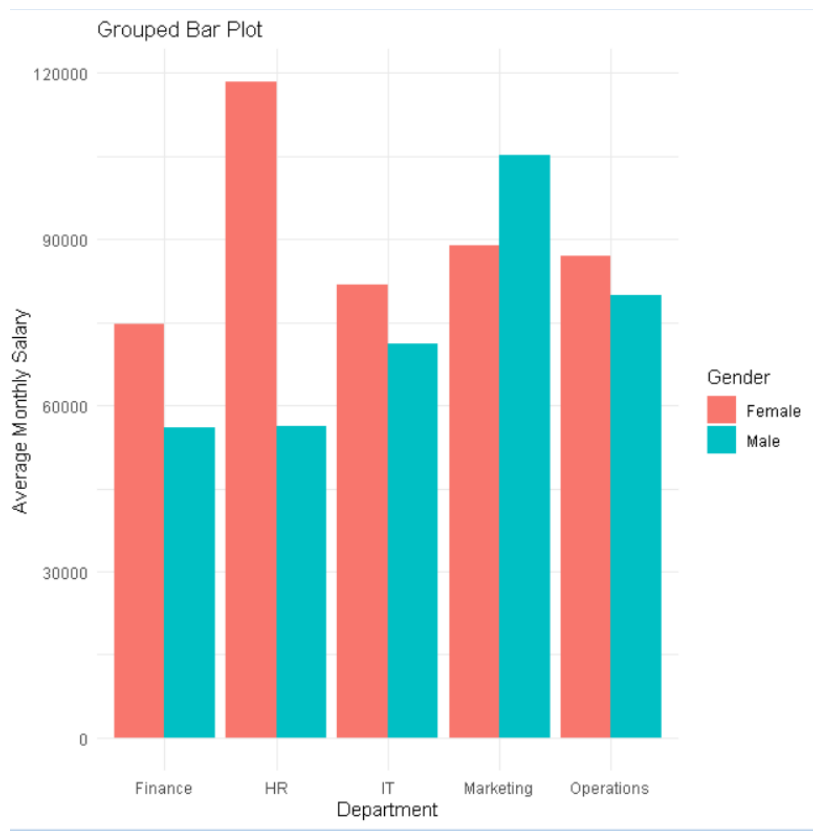
```
y = "Education Level") +  
theme_minimal()
```



## 4.GROUPED BAR PLOT

### R PROGRAM

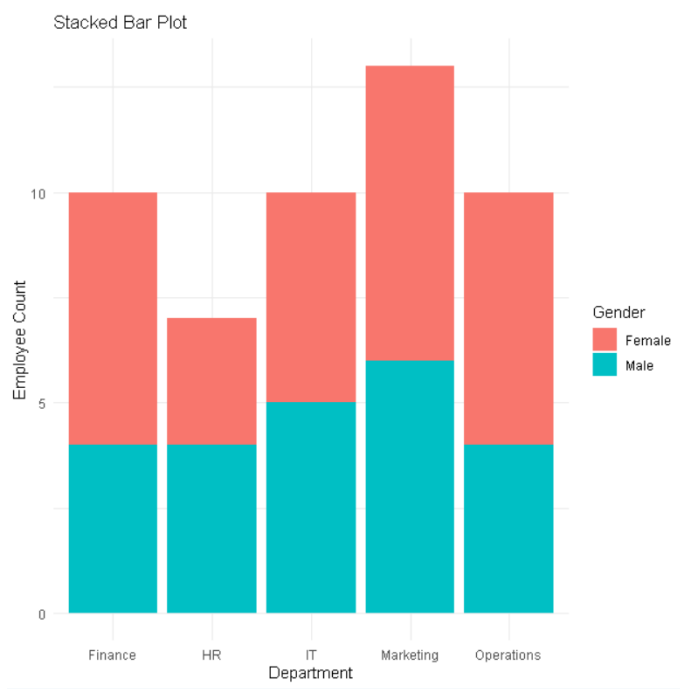
```
ggplot(employee,  
  aes(x = Department,  
    y = Monthly_Salary,  
    fill = Gender)) +  
stat_summary(fun = mean,  
  geom = "bar",  
  position = "dodge") +  
labs(title = "Grouped Bar Plot",  
  x = "Department",  
  y = "Average Monthly Salary") +  
theme_minimal()
```



## 5.STACKED BAR PLOT

### R PROGRAM:

```
ggplot(employee,  
  aes(x = Department,  
    fill = Gender)) +  
geom_bar() +  
labs(title = "Stacked Bar Plot",  
  x = "Department",  
  y = "Employee Count") +  
theme_minimal()
```

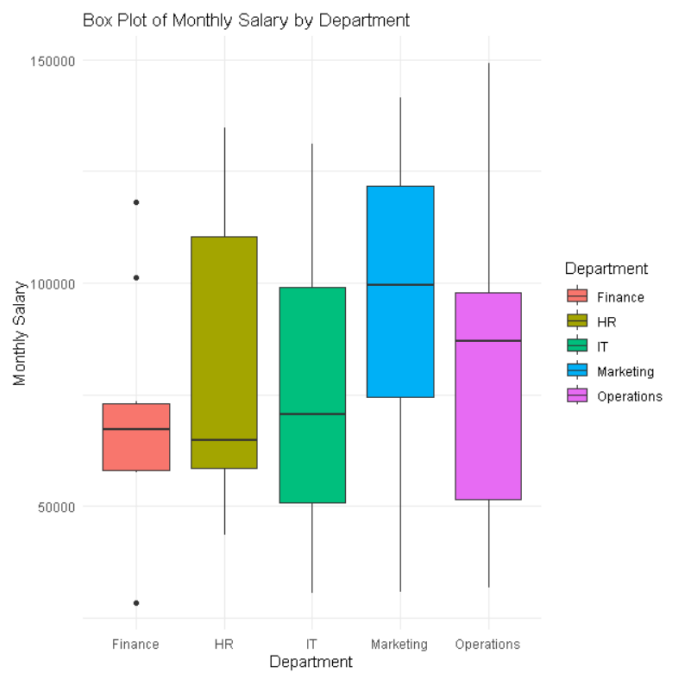


## 2.DISTRIBUTIONS

### 1.BOX PLOT VISUALIZATIONS

#### R PROGRAM

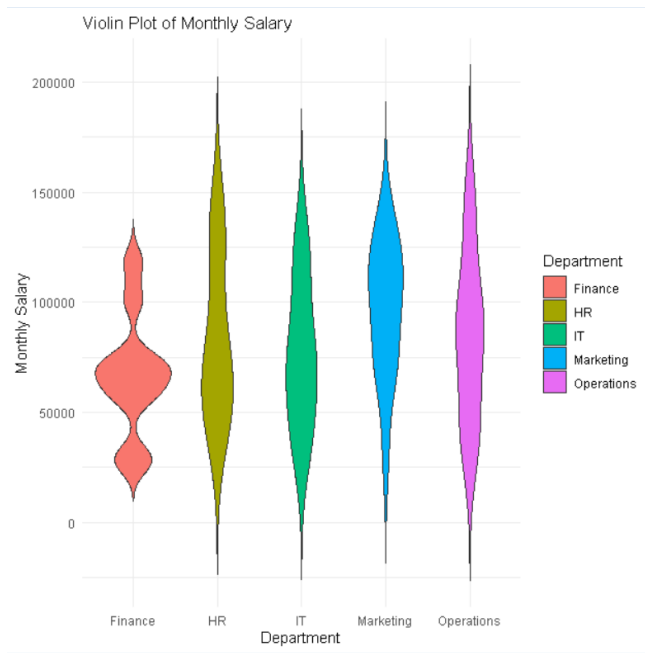
```
ggplot(employee,
  aes(x = Department,
    y = Monthly_Salary,
    fill = Department)) +
  geom_boxplot() +
  labs(title = "Box Plot of Monthly Salary by Department",
    x = "Department",
    y = "Monthly Salary") +
  theme_minimal()
```



## 2.VIOLIN PLOT VISUALIZATION

### R PROGRAM

```
ggplot(employee,
  aes(x = Department,
    y = Monthly_Salary,
    fill = Department)) +
  geom_violin(trim = FALSE) +
  labs(title = "Violin Plot of Monthly Salary",
    x = "Department",
    y = "Monthly Salary") +
  theme_minimal()
```



### 3.HISTOGRAM

#### R PROGRAM:

```
ggplot(employee,
  aes(x = Monthly_Salary)) +
  geom_histogram(binwidth = 10000,
    fill = "skyblue",
    color = "black") +
  labs(title = "Histogram of Monthly Salary",
    x = "Monthly Salary",
    y = "Frequency") +
  theme_minimal()
```



## 4.DENSITY PLOT

### R PROGRAM

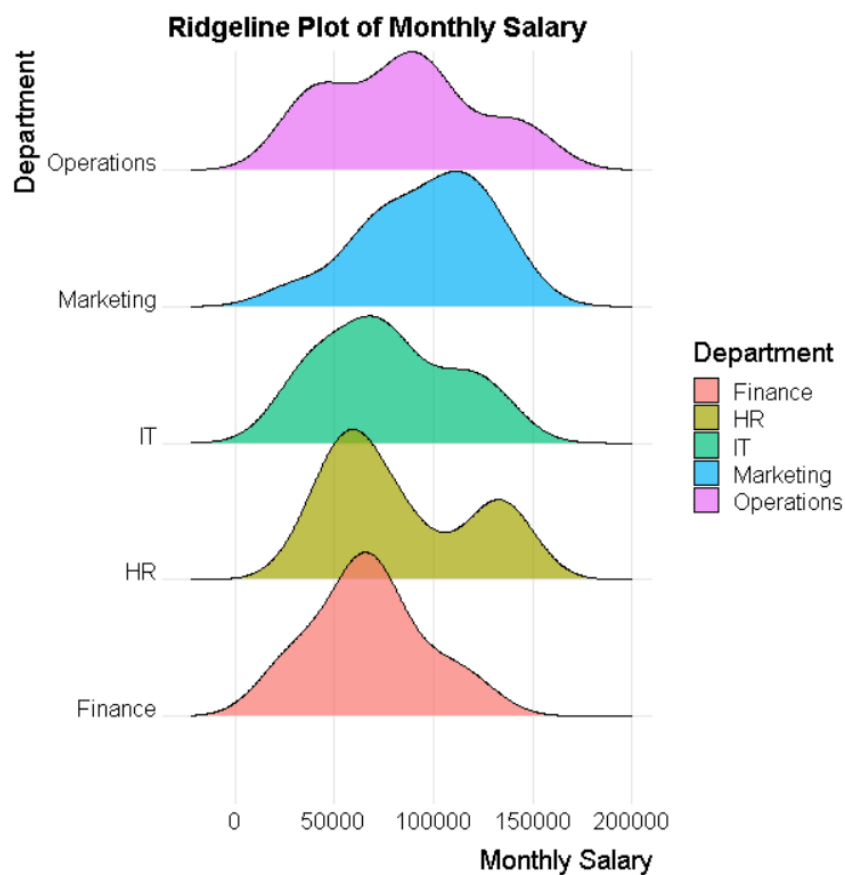
```
ggplot(employee,
  aes(x = Monthly_Salary,
    fill = Gender)) +
  geom_density(alpha = 0.5) +
  labs(title = "Density Plot of Monthly Salary",
    x = "Monthly Salary",
    y = "Density") +
  theme_minimal()
```



## 5. RIDGELINE (JOY) PLOT

## R PROGRAM:

```
ggplot(employee,  
  aes(x = Monthly_Salary,  
    y = Department,  
    fill = Department)) +  
geom_density_ridges(alpha = 0.7,  
  scale = 1.2) +  
labs(title = "Ridgeline Plot of Monthly Salary",  
  x = "Monthly Salary",  
  y = "Department") +  
theme_ridges()
```



---

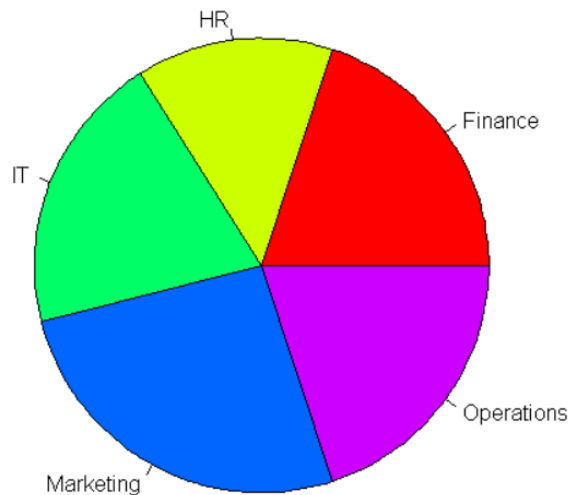
## 3.PROPORTIONS

### R PROGRAM

```
dept <- table(employee$Department)
```

```
pie(dept,
    col = rainbow(length(dept)),
    main = "Employee Distribution by Department")
```

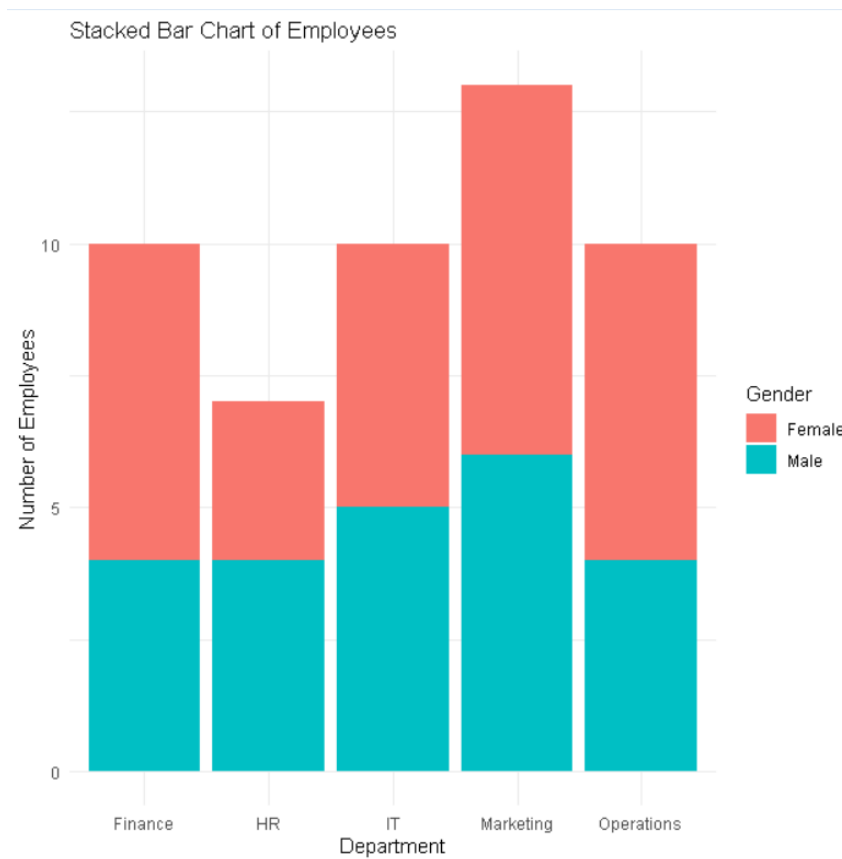
**Employee Distribution by Department**



## 2.STACKED BAR CHART

### R PROGRAM:

```
ggplot(employee,
    aes(x = Department,
        fill = Gender)) +
geom_bar() +
labs(title = "Stacked Bar Chart of Employees",
    x = "Department",
    y = "Number of Employees") +
theme_minimal()
```



### 3.TREE MAP

#### R PROGRAM

```
dept_data <- as.data.frame(table(employee$Department))  
colnames(dept_data) <- c("Department", "Count")  
treemap(  
  dept_data,  
  index = "Department",  
  vSize = "Count",  
  title = "Treemap of Employee Departments"  
)
```

Treemap of Employee Departments

